BEFORE YOU READ

Where permutation matrices live

Permutation matrices sit inside a chain of matrix classes: a few are elementary row swaps, all of them are orthogonal, and every orthogonal matrix is invertible.

$$E \subset P \subset Q \subset A$$

A — Invertible
A^{-1} exists

Q — Orthogonal
$Q^{-1} = Q^T$

P — Permutation — `n! of them`
adds any reordering · one 1 in each row and column

E — Row-swap elementary
swaps a single pair of rows

TAKEAWAY

Inverting is free

Because permutation matrices are orthogonal, $P^{-1} = P^T$: undoing a permutation means transposing it.